A SELF-AUTHORED USE CASE · SYNTHETIC DATA

Spotter

Agent assist for commercial lines service calls. Every regulated moment, caught as it happens.

Phil · October 2026

All companies, people, policies and figures are made up. Statutes and form numbers are real and cited.

THE CALL

Eleven minutes, three regulated moments, one client who called about a certificate

Marisol Reyes runs an 18-person contractor in Los Angeles. She needs a certificate of insurance by Monday. The contract asks for four things. Her policy carries two. Mid-call she mentions a new yard in Vernon and that she's started doing design work in-house.

MOMENT 1

The certificate

Completed-ops and primary wording aren't on the policy. A certificate that implies they are is the biggest E&O exposure in an agency.

MOMENT 2

The exposure

Design work is excluded by her GL. The product that covers it is placed in the non-admitted market, which triggers a statutory disclosure.

MOMENT 3

The words

The D-1 (Cal. Ins. Code §1764.1) is owned wording, signed at application. Paraphrasing it from memory is the violation.

THE PROBLEM

Today the account manager does it by tenure and tabs

The policy is in the AMS. The forms are PDFs. The carrier appetite guide is an email from March. The surplus lines disclosure is in one shared drive and the broker fee agreement is in another. What the client said at underwriting is in an application from two years ago.

No single view holds it. So the agent reads while the client waits, and the mistakes get caught by a file audit weeks later, on a sample.

#1

E&O exposure in an agency: a certificate showing coverage the policy doesn't give

12 to 18 mo

Ramp for a new account manager, and the knowledge was always in people's heads

0

People who catch the mistake during the call

THE USER

The commercial lines account manager is the one speaking, and the one liable

Why her

She holds the license. Her words about coverage are the regulated object. Producers and the compliance lead benefit downstream; the live call is hers.

The compliance pain

Three regulated moments per call in California alone: the recorded-line notice, the D-1 at a non-admitted quote, the broker fee disclosure. Each has owned wording.

The revenue she leaves on the call

Cross-sell dies on service calls because she's finishing the task the client called about. An agent with tenure hears "design work." One six months in doesn't.

Illustrative: if 1 in 5 service calls surfaces an uninsured exposure and 1 in 4 of those converts, a 2,000-call-a-month desk writes 100 placements a month it isn't writing today.

SERVICE DESIGN

The account manager stops reading and starts listening

Stage	Today	With Spotter
During, the work	Opens the AMS, the policy PDF, the carrier portal, the COI template. Listens with half an ear.	Reads the panel. Says what the forms match tells her, in her own words. Confirms what Spotter heard.
During, the E&O exposure	Issues a certificate off the contract instead of off the policy. Lets “design work” pass. Paraphrases the D-1 from memory.	Certificate locked while the gap exists. Exposure raised by name. D-1 on screen verbatim; quote blocked until it’s delivered.
After	Activity note from memory, often the next day. The file can’t defend the call.	Three drafts written from the event log. Review, edit, accept.
Who catches a mistake	QA file audit, weeks later, on a sample.	Spotter, before the words leave her mouth. Later: the same rules grade the transcript.
Handle time (illustrative)	18 min on the call, 6 on hold, 12 after	11 min on the call, 0 on hold, 3 after

WHAT SPOTTER DOES

Eleven beats. Four of them change the outcome of the call.

Beats 1 to 3

Recorded-line notice. Account card. Call type detected, four contract requirements parsed.

Beats 4 and 5

Forms read against the contract. Two missing. **Certificate blocked**, endorsement request offered.

Beats 6 and 7

Vernon yard caught, queued. Design work heard: GL excludes it, pro liability recommended, non-admitted.

Beats 8 and 9

D-1 and broker fee disclosure, verbatim. Exemption checked. **Quote blocked** until the D-1 is delivered.

Beats 10 and 11

Decline logged with a follow-up. Activity note, endorsement request, client email drafted for review.

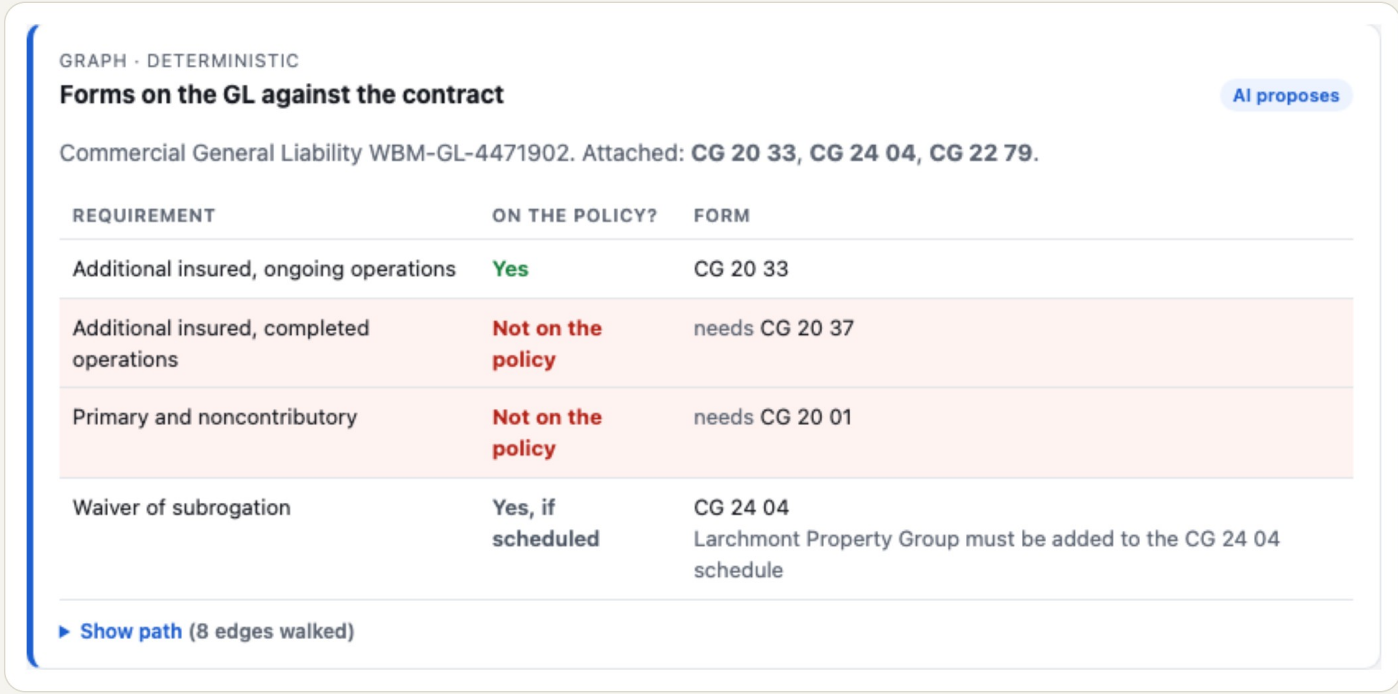
The transcript is scripted and says so on screen. The AMS is YAML. Model calls are recorded so the demo is deterministic. Each of those is labeled in the UI.

BEAT 4 · GROUNDED IN THE FORMS

Two of four requirements are on the policy

Spotter reads the endorsements actually attached to the GL (CG 20 33, CG 24 04, CG 22 79) and matches them to the contract's checklist. Completed ops and primary wording: not on the policy.

"Show path" opens the graph walk behind the answer. Eight edges. No model involved in this card.

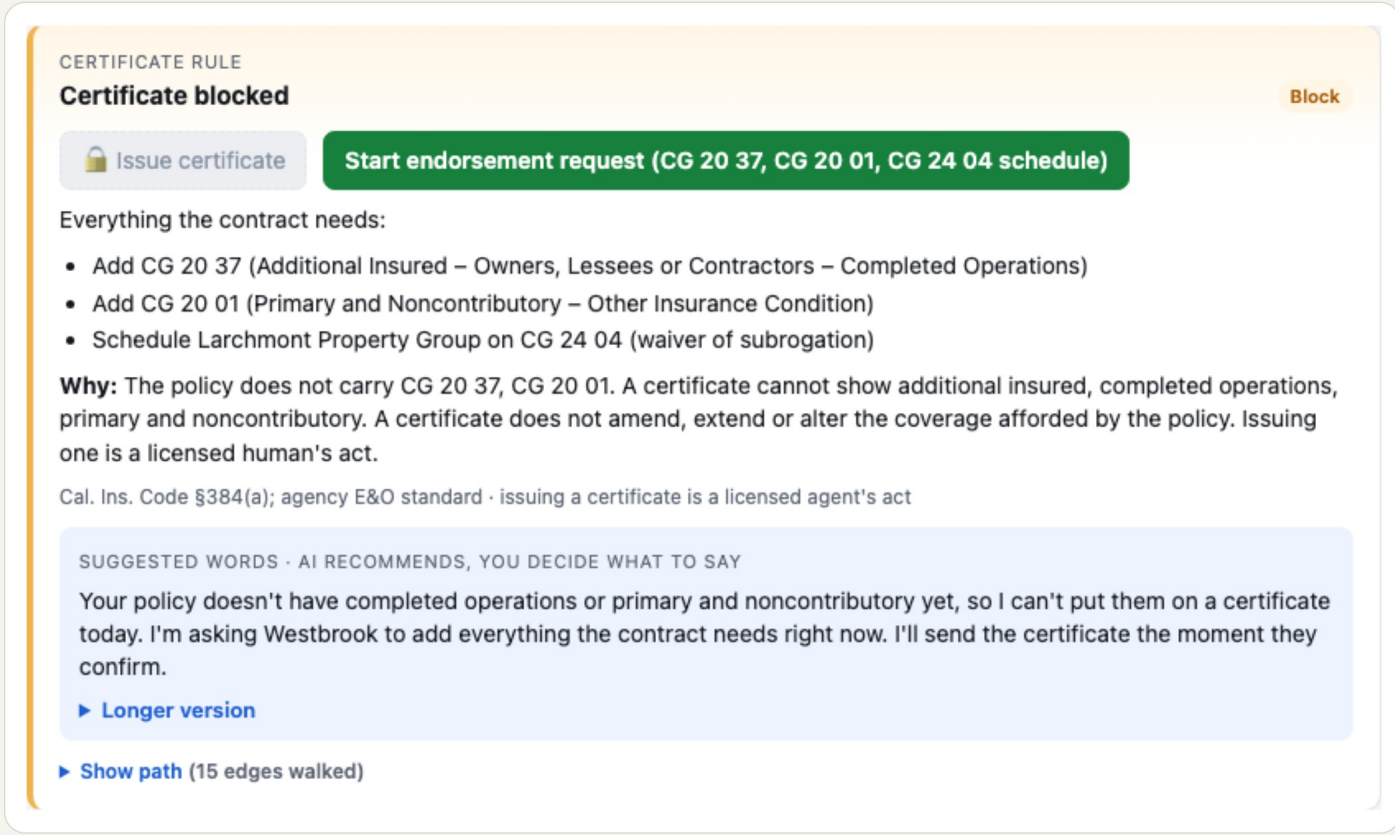


BEAT 5 · THE GAP, CAUGHT IN-FLOW

The certificate is locked before it can go out

“Issue certificate” is refused server-side with the reason and the citation. The green button offers the fix instead: an endorsement request naming every form the contract needs.

Suggested words are three sentences, so Dana can say them while Marisol waits. The longer version is one click away.

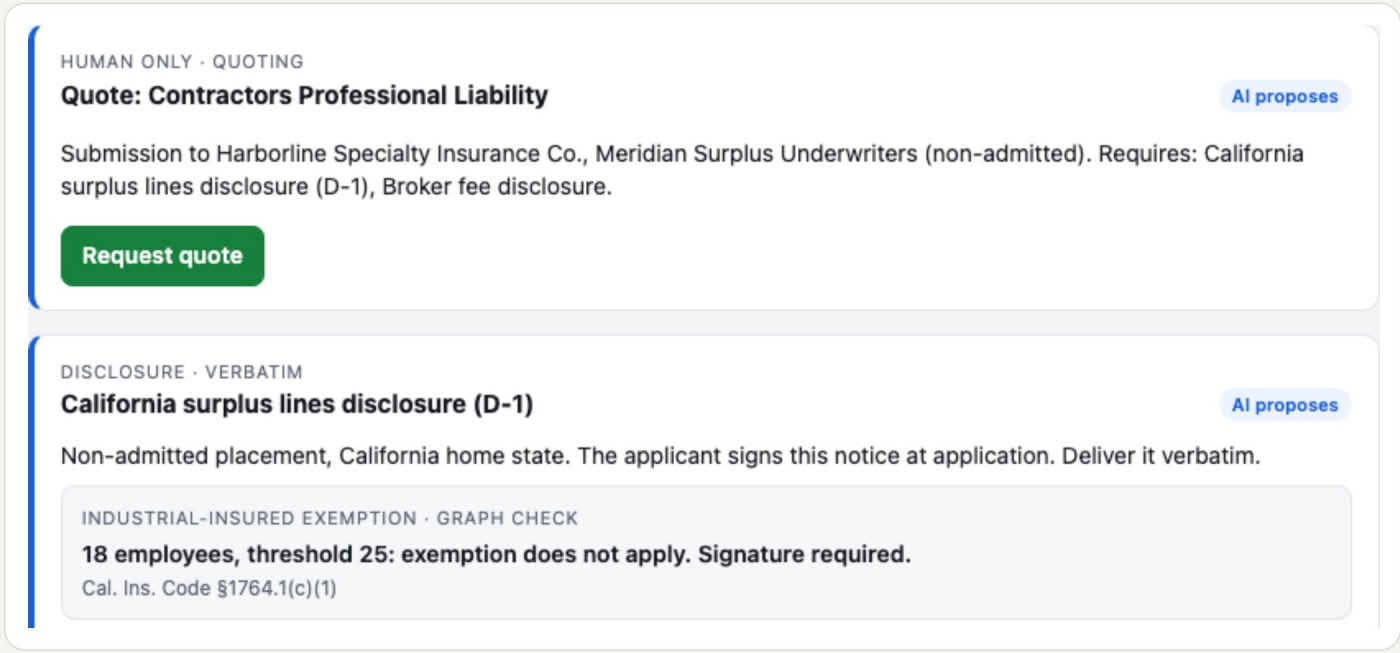


BEAT 8 · THE RIGHT WORDS AT THE RIGHT BEAT

The D-1 surfaces the moment the agent moves to quote

Contractors Professional Liability is placed non-admitted, so the graph selects the California surplus lines disclosure and the broker fee disclosure. Both verbatim, both cited.

The industrial-insured exemption is checked on screen: 18 employees, threshold 25, signature required. The app never writes or edits disclosure text; it reads the file.



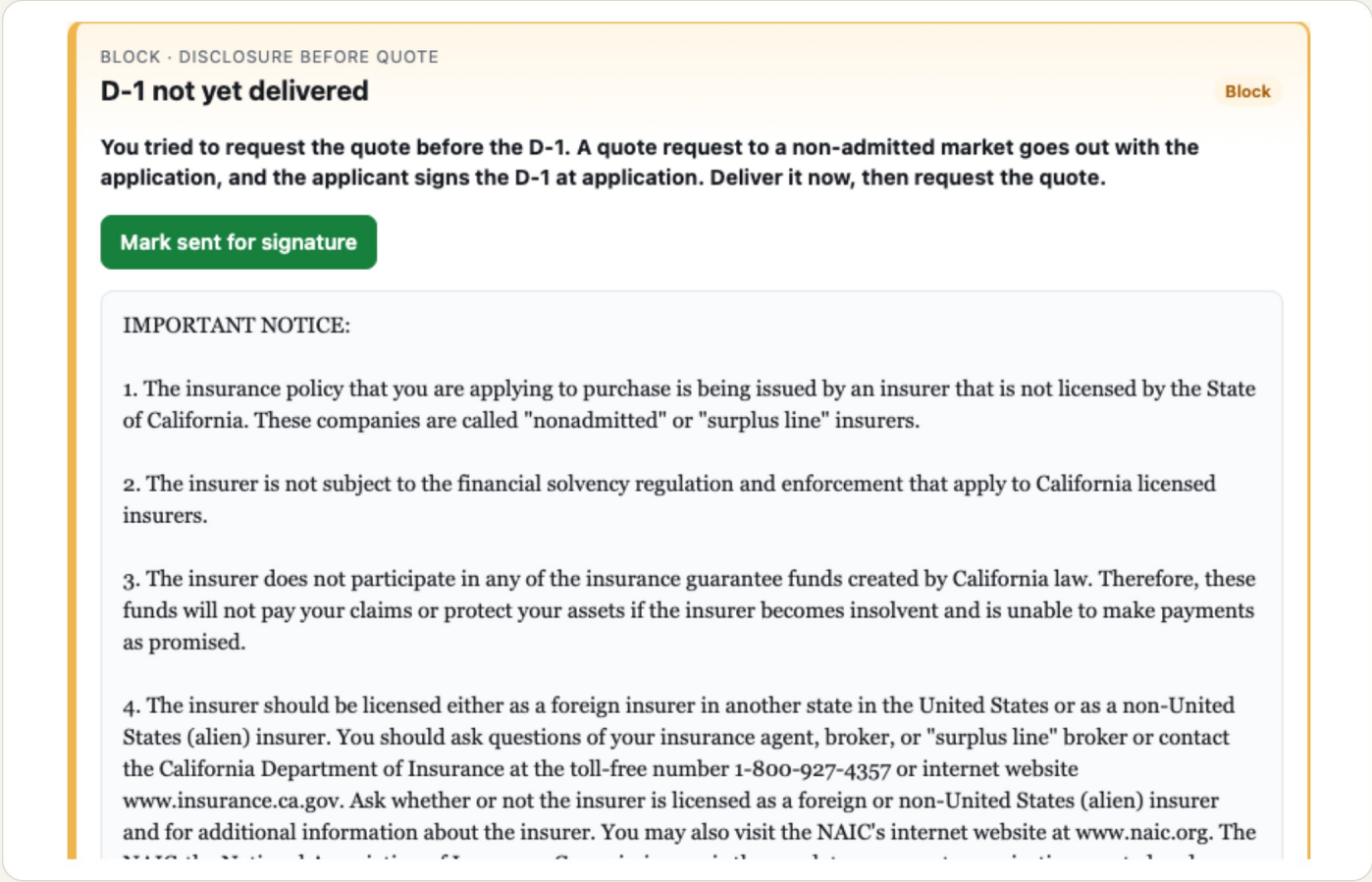
Below this card the full statutory notice is on screen, with “Mark sent for signature” underneath it.

BEAT 9 · THE NUDGE WHEN IT'S SKIPPED

Request the quote before the D-1 and the call stops

Marisol says "go ahead and get me a number." If Dana clicks "Request quote" before the D-1 is marked sent for signature, the amber block appears with the notice right there and the quote is refused.

A quote to a non-admitted market is treated as the application (§1764.1(a)(1)). An agency that gets indications first moves the same check to the application step: one line.

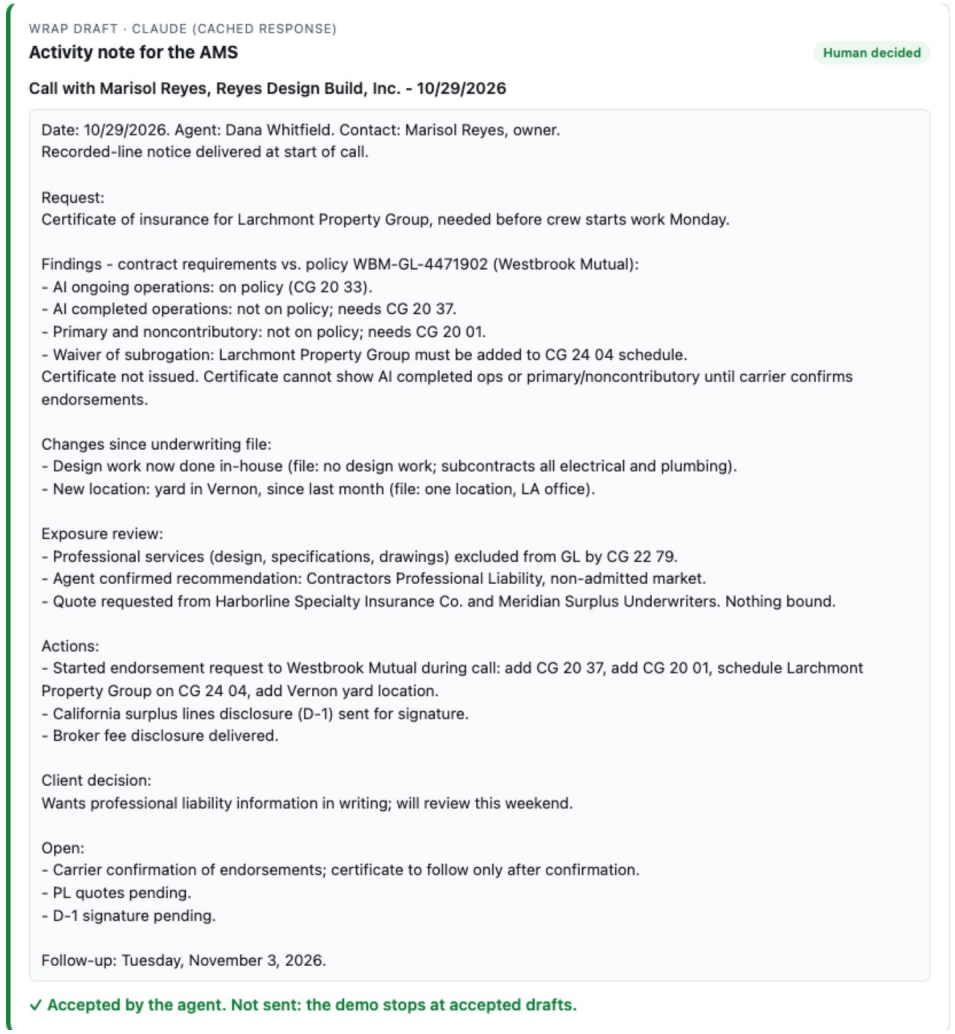


BEAT 11 · THE WRAP

Three drafts, written from the event log, blue until a human accepts

Activity note for the AMS. Endorsement request to Westbrook Mutual (CG 20 37, CG 20 01, the CG 24 04 schedule, the Vernon yard). Follow-up email to Marisol.

Claude drafts them from facts the graph already settled. The agent edits and accepts. Nothing is sent; the demo stops at accepted drafts and a logged click.



RESPONSIBLE AI

Three colors say who decided, on every card

BLUE · AI DECIDES

Detects call type and intent. Pulls the account and the forms. Detects a missed disclosure. Drafts the note, the endorsement request, the email.

Happens without a click. Detection is Claude; the account, forms match and disclosure selection are graph walks.

GREEN · HUMAN CONFIRMS

Which gaps to raise. Which product and which market. The next best question. Which endorsement answers the requirement. Disposition.

Shown blue with a button. Confirm, Dismiss, Start, Accept. Each click turns the card green and is logged with who made it.

AMBER · HUMAN ONLY

Represent coverage on a certificate. Quote or bind. Send a risk non-admitted. Deliver the signed disclosure. Agree a fee. Legal advice.

Locked or refused server-side with the reason. Disclosures are delivered and marked by the agent; the app never writes their text. Nothing binds, nothing is sent.

The brief's table, enforced in the UI rather than promised on a slide. Every proposal, block and decision lands in an append-only event log, which is what the after-call drafts and the future QA score read.

KNOWLEDGE ARCHITECTURE

The facts an agent needs are relationships, so the layer is a graph. I chose the hybrid.

	A · curated graph	B · LLM-extracted (GraphRAG)	C · hybrid (chosen)
Traceable path on every answer	Yes	Only if the extraction made the edge	Yes, on the curated edges; RAG citations for the text
Time to add a carrier's new form	Hours of hand modeling	Minutes	Minutes to propose, one human approval to accept
Who's accountable when it's wrong	The modeler	Nobody you can name	The approver, by design
Fit for a regulated call	Good, slow	Fast, not defensible	Good enough, and it extends

A small curated schema holds the edges that carry compliance meaning (form satisfies requirement, product placed in market, market triggers disclosure, exemption rule). Claude proposes edges from the documents into a review queue. A person approves anything compliance-bearing. Plain RAG over the statute and form text supplies the citation.

COMPETENCY QUESTIONS

Seven questions the graph has to answer with a path. All seven pass.

#	Question	Answer on the demo data
1	Does the GL give completed-ops additional insured status to a scheduled party?	No. CG 20 33 attached (ongoing only); CG 20 37 not attached.
2	Which of the contract's four requirements does the policy answer?	Ongoing AI yes. Waiver yes, if scheduled. Completed ops no. Primary wording no.
3	Which disclosures does Contractors Pro Liability trigger for a CA client with 18 employees?	D-1 (§1764.1) signed at application; broker fee disclosure. Exemption does not apply (18 < 25).
4	Which locations does the client operate at that the policy doesn't cover?	The Vernon yard, leased last month.
5	What changed since the underwriting file was written?	Design work now in-house. New location in Vernon.
6	Does the GL cover design work done in-house?	No (CG 22 79). Contractors Pro Liability, non-admitted.
7	Can a certificate showing completed-ops AI be issued today?	No. Human-only act, and CQ1 says the form isn't there. Route to endorsement.

10 tests across the seven questions; each asserts a non-empty path with named edges. 31 tests in the suite, 0 failing. Graph: 43 nodes, 48 edges, 21 flagged as compliance-bearing.

HOW IT'S BUILT

Content and workflow are separated. The call is one consumer of the layer.

KNOWLEDGE/

The governed source

Curated schema. Seed data for the demo world. Form, product and statute text. Verbatim disclosures, each with its citation and an owner.

→

GRAPH.PY · QUERIES.PY

The graph, deterministic

NetworkX, seeded from YAML. Every competency question returns its answer and the edges it walked. No model in this box.

→

ENGINE.PY · FASTAPI · HTMX

The call

Turns call state into cards. Forms match, certificate rule, disclosure choice, exemption check and quote block are deterministic. Append-only event log in SQLite.

Claude, in two places only (llm.py)

Per beat: intent and exposure detection, structured output limited to graph ids. At the wrap: the three drafts, from facts the graph settled. Responses recorded and replayed, so the demo needs no key.

extract.py: the other half of the hybrid

Claude reads each document and proposes edges against the schema. 15 documents, 32 proposals, 30 matching curated edges, 2 new. Nothing it proposes enters the graph until a person approves it.

THE KEY TRADEOFF

Compliance-bearing answers come from the graph, deterministically. The model only detects and drafts.

What it costs

A form the graph doesn't know produces "I can't tell you" instead of a plausible guess. A new carrier endorsement means someone approves an edge. On a demo day that looks slower than a model that answers everything.

Why I took it anyway

On a regulated call, the wrong confident answer is the E&O claim. A model reasoning over a 40-page form is right most of the time and fluent all of the time, and the agent can't tell which. A graph that says "no path" knows what it doesn't know.

PRODUCT ROADMAP

The layer has to exist before anything reads it, and be trusted before it's allowed near more calls

Phase 0 · built

The knowledge layer

One state, one line, five products, the forms that matter, three disclosures. Competency questions as tests. The asset everything else consumes.

Phase 1 · built

Live call assist

Spotter, this prototype. Reads the layer; writes nothing to it except the event log.

Phase 2 · review screen built, grading next

Approve the edges, then grade the call

The review screen for proposed edges is built: approve, reject, merge, with provenance. Next: score the event log against the same rules that guided the agent. Guided text equals graded text. This is where compliance signs off.

Phase 3

Renewal review and a second state

The renewal check-in reads the same layer for what changed. Texas is a scoping of edges by state; the call flow doesn't change, the disclosure node does.

Phase 4

Live audio and a real AMS

Speech-to-text and the AMS integration. Last on purpose: vendor work that changes none of the product's logic.

QA before renewal because QA is where compliance signs off. Integrations last because they're the expensive, slow, least differentiated part, and a prototype with scripted input already proves the product.

WHAT MODELING IT TAUGHT ME

Three places the statute is narrower than the brief reads. The graph made me look.

BROKER FEE · §1623

The statute requires the fee in a signed written agreement. It doesn't literally say "in advance." So the broker fee text is written as agency text citing §1623, and labeled that way.

CERTIFICATES · §384

§384 mandates the certificate disclaimer but carries no misrepresentation ban. The certificate rule cites §384(a) plus the agency's E&O standard, and says which is which.

EXEMPTION · §1764.1(C)

The industrial-insured exemption waives the signature only. The notice still goes on the confirmation or the policy. The graph models it that way; the brief's shorthand didn't.

A model reading the brief would have happily repeated the looser version in all three places. Having to write the edge down, with a citation on it, is what caught them.

THE THINKING BEHIND IT

What I didn’t build, what I assumed, and what I’d do next

Deliberately not built

- Pure GraphRAG. A day to ship; nobody can sign off an edge they never saw.
- Pure curated graph. Fine for 5 products; the 40th carrier form never gets modeled.
- Audio. A vendor call away, and it proves nothing about the layer.
- A chat box. Nothing on this screen waits for the agent to type.

Assumptions, stated

- The AMS exposes the policy, forms and underwriting file as structured data. Here it’s YAML.
- ISO form numbers are stable identifiers; a carrier’s own forms get their own nodes.
- One home state. Multi-state is edge scoping the schema already carries.
- The AI sees the transcript and the account. It can’t see carrier appetite, and doesn’t pretend to.

Next, and differently

- Next: grade the call with the same rules. Cheapest proof the layer is real.
- Then the approval queue on screen, so “a human approves the edge” is something you can watch.
- Differently: write the competency questions before the schema. Two entity types fell out of making CQ2 and CQ6 return a path.

SPOTTER

The system reads. The agent talks to the client.

A working eleven-beat call, 31 tests green, every grounded answer with a path, and a licensed human at every regulated moment.

Repo, README and the full write-up: github.com/phil-stt/clsc-agent-assist

All companies, people, policies and figures are made up.